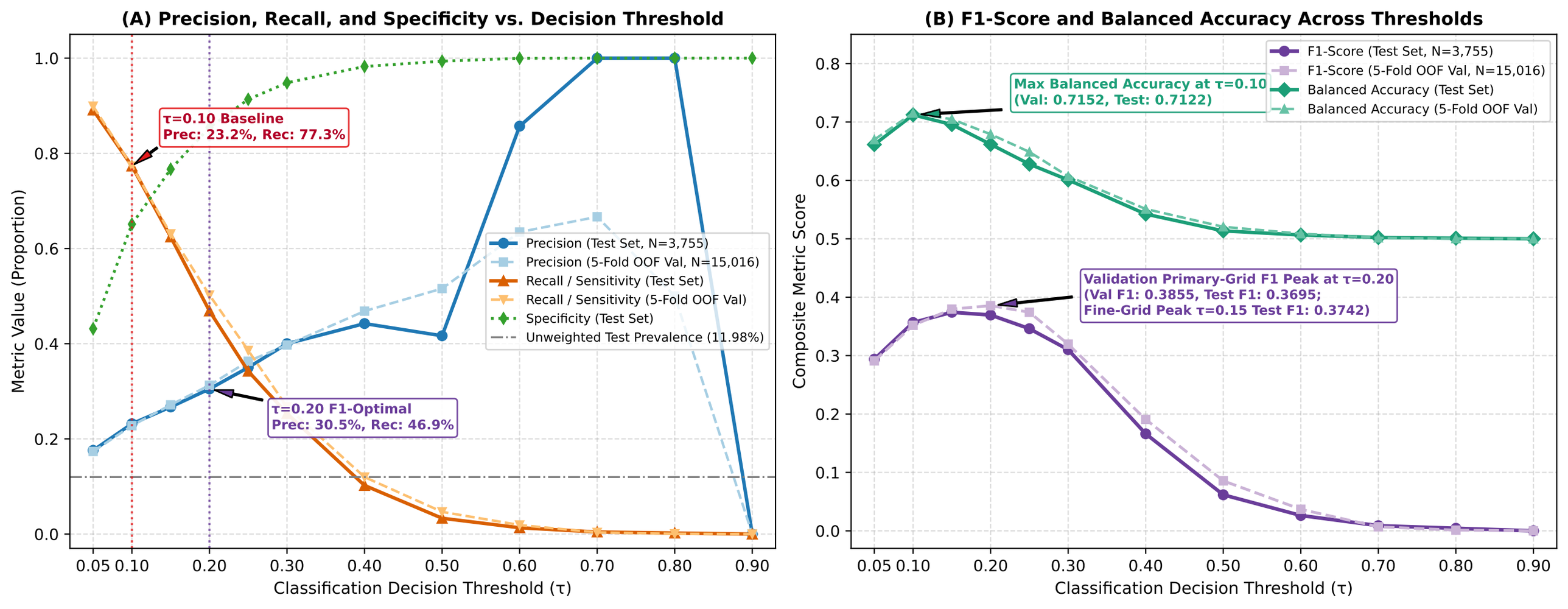
(A) Precision, Recall, and Specificity vs. Decision Threshold
Metric Value (Proportion)
Classification Decision Threshold (τ)
τ=0.10 Baseline
Prec: 23.2%, Rec: 77.3%
τ=0.20 F1-Optimal
Prec: 30.5%, Rec: 46.9%
Precision (Test Set, N=3,755)
Precision (5-Fold OOF Val, N=15,016)
Recall / Sensitivity (Test Set)
Recall / Sensitivity (5-Fold OOF Val)
Specificity (Test Set)
Unweighted Test Prevalence (11.98%)
(B) F1-Score and Balanced Accuracy Across Thresholds
Composite Metric Score
Classification Decision Threshold (τ)
Max Balanced Accuracy at τ=0.10
(Val: 0.7152, Test: 0.7122)
Validation Primary-Grid F1 Peak at τ=0.20
(Val F1: 0.3855, Test F1: 0.3695;
Fine-Grid Peak τ=0.15 Test F1: 0.3742)
F1-Score (Test Set, N=3,755)
F1-Score (5-Fold OOF Val, N=15,016)
Balanced Accuracy (Test Set)
Balanced Accuracy (5-Fold OOF Val)